

Software Testing Documentation

1■■■ What is Software Testing?

Software Testing ka matlab hota hai software ko check karna ki wo sahi kaam kar raha hai ya nahi, matlab usme bugs (errors) to nahi hain aur wo expected output de raha hai ya nahi. Testing ka main goal hai ensure karna ki end user ko ek perfect aur reliable product mile.

2■■■ Purpose / Uses of Testing

- Software ke bugs aur errors find karna
- System ke functionality verify karna
- Software requirement ke hisaab se bana hai ya nahi check karna
- User experience ensure karna
- Cost aur time save karna (early testing se later bug fixing mehenga nahi padta)

3■■■ Types of Software Testing

Software Testing do main parts me divide hoti hai: Manual Testing aur Automation Testing.

■ *A. Manual Testing*

- Functional Testing – check karta hai ki function sahi kaam kar rahe hain (like login, signup).
- Regression Testing – purane features abhi bhi sahi chal rahe hain ya nahi new code ke baad.
- Smoke Testing – basic build test hota hai (software run ho raha hai ya crash kar raha hai?).
- Sanity Testing – quick check after minor code change.
- User Acceptance Testing (UAT) – end user check karta hai ki system use karne layak hai ya nahi.
- Exploratory Testing – bina predefined test cases ke tester manually explore karta hai system.

■ *B. Automation Testing*

- Unit Testing – code ke smallest parts test karte hain.
- Integration Testing – modules ek saath properly kaam kar rahe hain ya nahi.
- System Testing – poore system ko test karte hain end-to-end.
- Performance Testing – system ki speed aur stability check karte hain.
- Security Testing – vulnerabilities aur data protection verify karte hain.
- API Testing – backend APIs sahi response de rahe hain ya nahi check karte hain.
- Regression Testing (automated) – automated tools ke through pura regression suite run hota hai.

4■■■ Popular Testing Tools

Different testing purposes ke liye commonly used tools niche diye gaye hain:

- Manual Testing: JIRA, Bugzilla, Trello, TestLink, Zephyr.
- Automation Testing: Selenium, Cypress, Playwright, TestComplete, Katalon Studio.
- API Testing: Postman, SoapUI, RestAssured.
- Performance Testing: JMeter, LoadRunner, Gatling.
- Security Testing: OWASP ZAP, Burp Suite, Nessus.

5■■■ Real-World Example

Example: Tum ek E-commerce website (like Flipkart or Amazon) test kar rahe ho. Testing steps honge: 1. Functional Testing → Add to Cart, Payment Gateway, Login test karna. 2. Regression Testing → New feature add hone ke baad purane kaam sahi chal rahe hain? 3. Performance Testing → 10,000 log ek saath site kholte hain to site slow to nahi ho rahi? 4. Security Testing → Payment aur user data secure hai ya nahi? 5. Compatibility Testing → Website har browser aur device pe sahi chal rahi hai ya nahi?

6■■■ Career in Software Testing

- QA Tester / Engineer
- Automation Test Engineer
- Test Analyst
- Performance Tester
- SDET (Software Development Engineer in Test)

Languages: Python, Java, JavaScript seekhne se automation testing me growth hoti hai.

7■■■ Summary

Software Testing ensure karta hai ki software bug-free, secure aur user-friendly ho. Manual aur automation tools dono ka use karke testing ki jaati hai taaki high-quality product deliver ho.